

MQP SOLUTIONS (BME654B)

MODULE 01

1a. Explain the need of non-conventional energy sources.

1. Depletion of Fossil Fuels:

- Conventional sources like coal, oil, and natural gas are limited and depleting fast.
- Non-conventional sources (solar, wind, etc.) are renewable and inexhaustible.

2. Environmental Pollution:

- Fossil fuel usage leads to CO₂ emissions, air and water pollution.
- Non-conventional sources are clean, eco-friendly, and reduce climate change impacts.

3. Energy Security:

- Heavy dependence on oil imports affects national energy security.
- Using local renewable sources ensures energy independence.

4. Growing Energy Demand:

- Increasing population and industrialization have caused high energy demand.
- Non-conventional sources help meet this demand sustainably.

5. Rural and Remote Area Access:

- Non-conventional sources like solar and wind can provide energy in areas where grid connection is not feasible.

6. Sustainable Development:

- These sources support long-term development without harming the environment or exhausting resources.

7. Government Support:

- Encouraged by policies, subsidies, and international agreements for clean energy transition.

1b. List the advantages and disadvantages of renewable energy resources.

Advantages:

1. **Environment friendly** – Produces little or no pollution.
2. **Sustainable** – Never-ending sources like sun, wind, water.
3. **Reduces import dependency** – Uses locally available resources.
4. **Low operating cost** – Once installed, running cost is low.
5. **Job creation** – Boosts green employment in new sectors.

Disadvantages:

1. **High initial cost** – Expensive setup for solar, wind systems.
2. **Intermittent supply** – Depends on weather (sun, wind).
3. **Storage needed** – Requires battery systems for reliability.
4. **Large land requirement** – Big wind/solar farms need more space.
5. **Technological dependency** – Needs smart grids and modern infra.

Conclusion:

- **Renewables** are clean and sustainable but need better technology and investment.
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2a. With schematic representation explain mechanism of absorption, scattering beam and diffused radiation received at Earth's surface

1. Absorption:

- Some portion of solar radiation is absorbed by atmospheric gases like ozone (O_3), CO_2 , H_2O vapor, and dust.
- This absorbed energy heats the atmosphere.
- Radiation in UV range is mostly absorbed before reaching the Earth's surface.

2. Scattering:

- When solar radiation hits air molecules, water droplets, or dust particles, it gets scattered in various directions.

Two main types:

- **Rayleigh Scattering:** Due to molecules (affects shorter wavelengths like blue → blue sky)
- **Mie Scattering:** Due to larger particles (affects all wavelengths → white clouds)

3. Beam (Direct) Radiation:

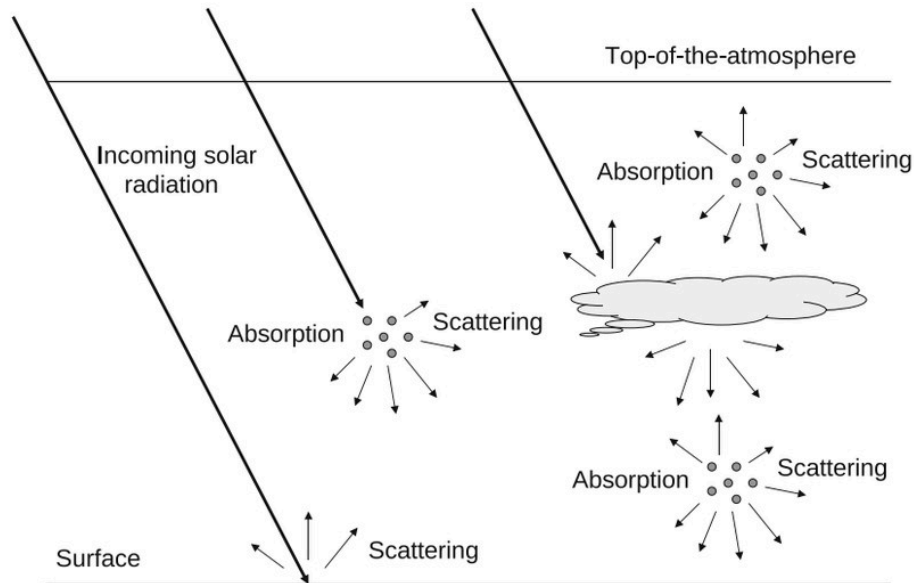
- The portion of solar radiation that reaches the Earth without any deviation.
- Travels in a straight line from the Sun to the surface.
- Useful for solar concentrators and PV panels with sun-tracking systems.

4. Diffused Radiation:

- Radiation that has been scattered by the atmosphere but still reaches the Earth.
- Non-directional – comes from all over the sky.
- Important in cloudy conditions and for fixed flat-plate solar collectors.

5. Global Solar Radiation:

- The total solar radiation received on a horizontal surface.
- It is the sum of beam and diffused radiation.



Conclusion:

Solar radiation reaching Earth undergoes absorption and scattering, and reaches the surface as beam (direct) and diffused components. Understanding this mechanism helps in designing efficient solar energy systems.

2b. Explain spectral distribution of extra-terrestrial radiation.

- Spectral distribution refers to the variation of solar irradiance with wavelength of solar radiation as it is received at the top of Earth's atmosphere.
- The solar spectrum ranges from $0.1 \mu\text{m}$ to about $3 \mu\text{m}$, and is divided into five key regions:

1. Ultraviolet C (UVC) Range:

- **Wavelength:** $0.1 \mu\text{m}$ to $0.28 \mu\text{m}$
- High-frequency radiation, **invisible to human eyes**
- **Absorbed by atmosphere**, only tiny amount reaches Earth

2. Ultraviolet B (UVB) Range:

- **Wavelength:** 0.28 μm to 0.315 μm
- **Also absorbed** by atmosphere
- Along with UVC, responsible for **ozone layer production**

3. Ultraviolet A (UVA) Range:

- **Wavelength:** 0.315 μm to 0.4 μm
- Less damaging to DNA
- Used in **tanning** and **PUVA therapy** (for skin conditions like psoriasis)

4. Visible Range (Light):

- **Wavelength:** 0.38 μm to 0.78 μm
- This is the **only part visible to human eye**
- Important for **photosynthesis** and **solar power conversion**

5. Infrared (IR) Range:

- **Wavelength:** 0.7 μm to 100 μm
 - Major contributor to **heat energy on Earth**
 - Divided into:
 - **Infrared-A:** 0.7 – 1.4 μm
 - **Infrared-B:** 1.4 – 3.0 μm
 - **Infrared-C:** 3.0 – 100 μm
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MODULE 02

3a. Explain with a neat sketch working of pyranometer.

1. Purpose:

- Pyranometer is used to measure global solar radiation (i.e., sum of beam and diffuse radiation) falling on a horizontal surface.

2. Principle of Operation:

- It works on the thermopile principle, where temperature difference between hot and cold junctions produces a voltage.
- This voltage is proportional to the solar radiation intensity.

3. Construction:

- **Black surface (sensor):** Absorbs all solar radiation.
- **White surface (reference):** Reflects radiation and stays cooler.
- **Thermocouples:** Measure temperature difference between black and white surfaces.
- **Glass dome:** Protects sensor and allows solar radiation to enter.
- **Level base:** Ensures proper horizontal positioning for accuracy.

4. Working:

- Sunlight enters through the glass dome and strikes the black coated surface.
- The black surface absorbs radiation and becomes hotter than the white surface.
- This causes a temperature difference, which generates a small voltage in the thermocouples.
- This voltage is calibrated and displayed as radiation intensity (W/m^2).

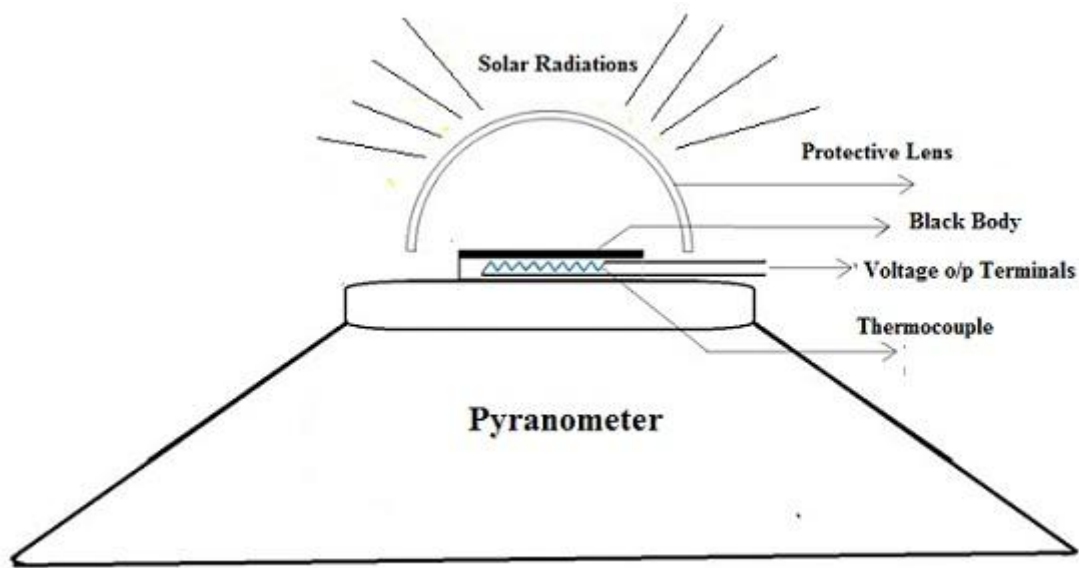


Fig : Pyranometer

3b. With a neat sketch explain working of liquid flat plate collector.

1. Purpose:

- Used to collect solar energy and convert it into heat, which is then transferred to a liquid (like water or antifreeze) flowing through tubes.

2. Principle:

- It works on the principle of absorbing solar radiation on a black surface, converting it to heat energy, and then transferring the heat to a liquid medium through conduction.

3. Construction:

- **Absorber Plate:** A **black-coated metal sheet** (usually copper or aluminum) to absorb solar radiation.
- **Transparent Cover (Glazing):** A **glass sheet** that allows sunlight in but reduces **heat loss by convection and radiation**.
- **Insulation:** Provided at the bottom and sides to **minimize heat losses**.

- **Tubes or Fins:** Carry the **liquid (water/antifreeze)** in contact with the absorber to collect heat.
- **Casing/Box:**
Encloses all components and protects from environmental damage.

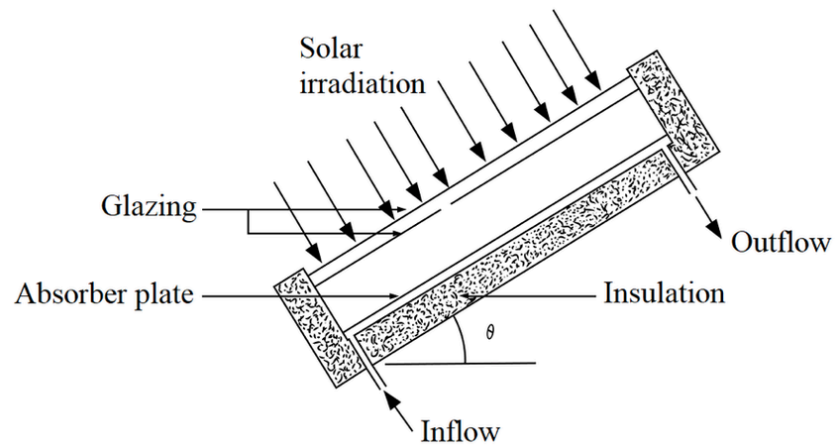


Fig : flat plate collector

4. Working:

- Sunlight passes through the glass cover and is absorbed by the black absorber plate.
 - The absorber heats up and transfers heat to the liquid flowing in the tubes.
 - The heated liquid is carried to a storage tank for use.
 - The system is either open loop (direct heating) or closed loop (with heat exchanger).
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4a. Explain the working principle and I-C Characteristics of Solar PV Cell.

Working Principle of Solar PV Cell:

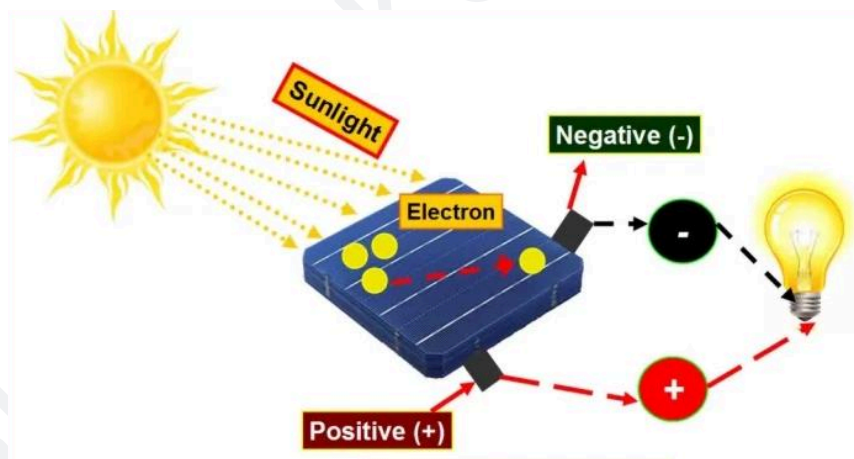
A solar photovoltaic (PV) cell works on the principle of the photovoltaic effect, where sunlight is directly converted into electricity using semiconductor materials like silicon.

1. Light Absorption: When sunlight falls on the surface of the PV cell, photons are absorbed by the semiconductor material.

2. Electron-Hole Pair Generation: The energy from the photons excites electrons, causing them to break free from atoms and create electron-hole pairs.

3. Separation of Charges: Due to the built-in electric field in the p-n junction, electrons move towards the n-side and holes move towards the p-side.

4. Current Flow: When the circuit is connected externally, electrons flow through the load, generating direct current (DC) electricity.



I-V Characteristics of Solar PV Cell:

The I-V (**Current vs Voltage**) characteristics show how the output current varies with voltage under constant solar radiation.

1. Short Circuit Current (I_{sc}): The current when output terminals are shorted ($V = 0$). It is the maximum current the cell can deliver.

2. Open Circuit Voltage (Voc): The voltage when no load is connected ($I = 0$). It is the maximum voltage of the cell.

3. Maximum Power Point (MPP): The point on the I–V curve where the product of current $\times$ voltage is maximum. It defines the cell's best operating condition.

4. Fill Factor (FF):

Defined as:

$$FF = (V_{mp} \times I_{mp}) / (V_{oc} \times I_{sc})$$

It indicates the quality and efficiency of the solar cell.

5. Effect of Illumination: As solar radiation increases, both current and power output of the cell increase.

Conclusion:

Solar PV cells convert sunlight into electricity using the photovoltaic effect. The I–V characteristics help to analyze the **performance, efficiency, and behavior** of the solar cell under different conditions.

4b. Discuss how regular operation, maintenance, and performance monitoring contribute to the long-term reliability of a solar power plant.

1. Regular Operation Ensures System Stability

- Proper day-to-day operation ensures that all components function as intended.
- Avoids unexpected shutdowns and keeps power generation stable.

2. Scheduled Maintenance Prevents Failures

- Routine maintenance includes cleaning panels, checking connections, inverter servicing, etc.
- It helps in identifying and fixing minor issues before they become major faults.

3. Panel Cleaning Increases Efficiency

- Dust, dirt, and bird droppings reduce sunlight absorption.
- Regular cleaning ensures maximum energy capture and improves efficiency.

4. Monitoring System Performance Helps Optimization

- Real-time monitoring of output (voltage, current, power) helps detect underperformance.
- Deviations from expected performance can indicate shading, equipment failure, or wiring issues.

5. Fault Detection and Quick Repair

- Automated alerts help in early detection of faults like inverter trip, grid disconnect, etc.
- Faster response reduces downtime and loss in energy production.

6. Data Logging Supports Long-Term Planning

- Storing performance data helps analyze trends, degradation, and predict maintenance needs.
 - Useful for system upgrades, expansions, and financial planning.
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MODULE 03

5a. List types of wind mills. Explain Horizontal Axis Wind Machine.

Types of Wind Mills: Wind machines are broadly classified into two types based on the axis of rotation:

1. Horizontal Axis Wind Machines (HAWM):

- Rotor shaft is horizontal to the ground and aligned parallel to wind direction.
- Most commonly used for electricity generation.

2. Vertical Axis Wind Machines (VAWM):

- Rotor shaft is perpendicular to the ground.
- Can accept wind from any direction, but generally less efficient.

Horizontal Axis Wind Machine

1. Orientation and Design:

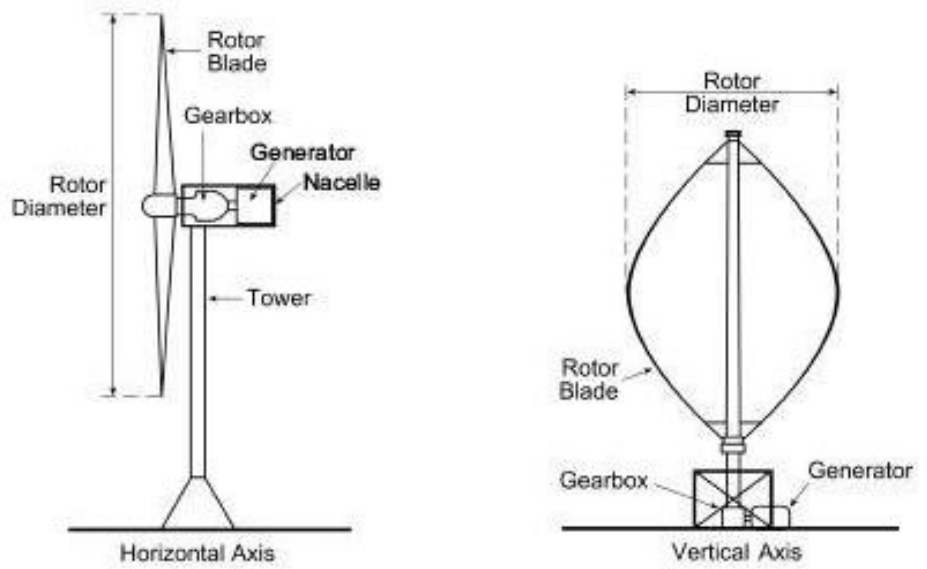
- The rotor axis is horizontal, and blades rotate in a vertical plane.
- The machine must be oriented to face the wind using a yaw mechanism.

2. Main Components:

- **Rotor Blades:** Capture wind energy through **aerodynamic lift**.
- **Nacelle:** Houses the **gearbox and generator**.
- **Gearbox:** Converts low-speed rotation from rotor to **high-speed rotation** for the generator.
- **Generator:** Converts **mechanical energy into electrical energy**.
- **Yaw System:** Keeps the rotor facing the wind.
- **Tower:** Elevates the turbine to reach **stronger, steadier wind** at higher altitudes.

3. Working Principle:

- Wind hits the rotor blades → creates lift force → blades rotate.
- This rotation drives the low-speed shaft, which connects to the gearbox.
- The gearbox increases the speed and turns the high-speed shaft, which drives the generator.
- Generated electricity is then either used locally or fed to the grid.



4. Advantages:

- Higher efficiency due to precise alignment with wind direction.
- Well-suited for utility-scale applications and wind farms.
- Advanced control systems provide better performance in varying wind conditions.

5. Limitations:

- Requires yaw mechanism to track wind direction.
- Installation and maintenance at higher altitudes can be challenging.

5b. Describe the main consideration in selecting the site for wind generator.

Main Considerations for Selecting the Site for Wind Generator:

1. Wind Speed and Consistency

- Site should have sufficient and steady wind speed, ideally above 5 m/s.
- More wind speed = more power generation.

2. Wind Direction

- Dominant wind direction should be known and stable.
- Helps in optimizing turbine orientation for maximum efficiency.

3. Topography and Terrain

- Elevated and open landscapes (like hills, coastal areas) are preferred.
- Avoid valleys, forests, or obstacle-rich areas that cause turbulence or shadowing.

4. Land Availability and Accessibility

- Enough flat land required for installation and maintenance.
- Site should be easily accessible for transport of components and servicing.

5. Distance from Grid Connection Point

- The site should be reasonably close to the power grid to reduce transmission losses and cost.

6. Environmental and Social Impact

- Should not disturb wildlife, local habitats, or human settlements.
- Noise and visual impact should be within acceptable limits.

7. Soil and Foundation Conditions

- The soil must support the foundation and tower weight.
- Poor soil may need special engineering → increases cost.

6a. Describe construction and working principle of geothermal energy with schematic diagram.

Geothermal Energy – Construction and Working Principle

1. Purpose:

- Geothermal energy is the heat energy stored beneath the Earth's surface, which can be used for electricity generation and heating applications.

2. Construction of a Basic Geothermal Power Plant:

- **Production Well:** A deep well is drilled into geothermal reservoirs to extract hot water or steam.
- **Steam Separator / Flash Chamber (if needed):** If hot water is extracted, it is partially flashed into steam using a pressure-reducing chamber.
- **Steam Turbine:** The steam is sent to a turbine, where it rotates the blades.
- **Generator:** The turbine is connected to a generator, which converts the mechanical energy into electricity.
- **Condenser:** After doing work, the used steam is condensed into water for reuse or reinjection.
- **Cooling Tower (Optional):** Removes excess heat from the condensed water.
- **Reinjection Well:** The cooled water is pumped back into the Earth to maintain reservoir pressure and sustainability.

3. Working Principle:

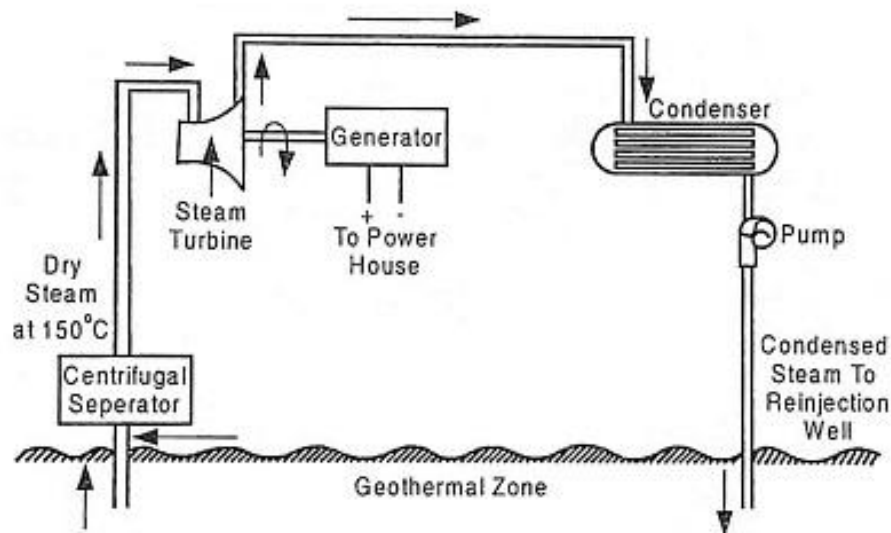


Fig. 4.48 Dry Steam Open System

1. Heat energy from the Earth's core heats underground water and forms hot water or steam in natural reservoirs.
2. Through production wells, this heat in the form of steam or hot water is brought to the surface.
3. If steam is available directly (dry steam system), it goes straight to the turbine.
4. If hot water is extracted (flash steam system), it is flashed into steam using a flash tank, then sent to the turbine.
The steam rotates the turbine, which is coupled to a generator, producing electricity.
5. Used steam is cooled, condensed, and reinjected into the reservoir, completing the closed-loop cycle.

6b. List and explain the problems associated with Geothermal System operations.

1. Depletion of Geothermal Reservoirs

- Continuous extraction of heat and fluid can lead to pressure drop and reduced steam production over time.

2. Land Subsidence

- Withdrawal of underground fluids may cause the land above to sink or collapse, especially if reinjection is not done properly.

3. Corrosion and Scaling

- Geothermal fluids often contain dissolved minerals and gases which cause corrosion of pipes and scaling on turbine blades and heat exchangers.

4. Environmental Pollution

- Although cleaner than fossil fuels, geothermal systems may release toxic gases like hydrogen sulfide (H_2S), arsenic, and boron, which can be harmful if not controlled.

5. Noise Pollution

- Drilling operations and turbine machinery may produce significant noise, especially during the initial stages of plant setup.

6. High Initial Cost

- Exploration, drilling, and setup require high investment, and site risk is involved (steam may not be found in expected quantity).

7. Earthquakes (Seismic Risk)

- Reinjection of water into geothermal wells can cause minor seismic activities due to pressure changes underground.

8. Limited Site Availability

- Suitable geothermal sites are limited to areas with natural geothermal activity, like volcanic zones or tectonic plate boundaries.

MODULE 04

7a. Explain with a sketch double basin tidal power plant. .

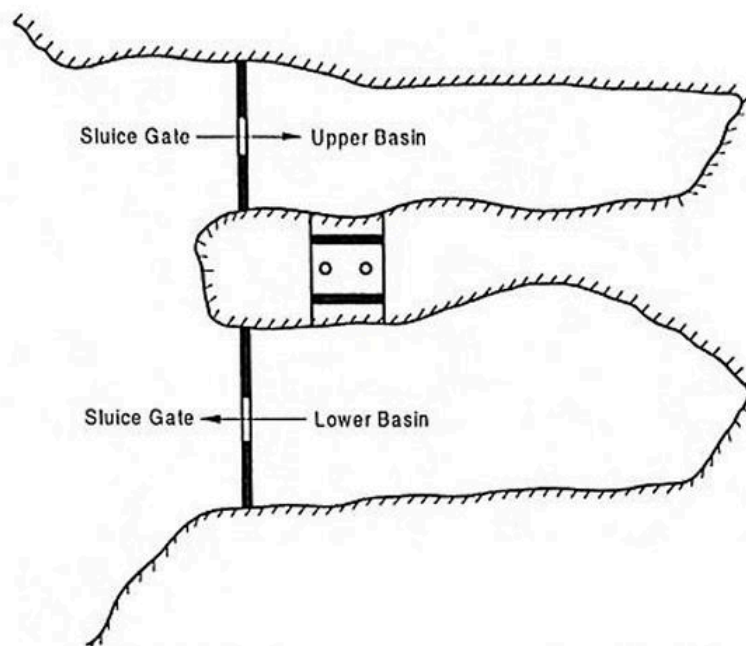
1. What is a Double Basin System?

- A double basin tidal power system uses two basins at different levels to enable continuous or near-continuous power generation.
- One basin stores water during high tide, and the other discharges it during low tide, allowing controlled flow through turbines.

2. Construction Overview:

- **Consists of:**
 - **Upper basin** (closer to the sea)
 - **Lower basin** (closer to land)
 - **Barrages/dams** to separate basins
 - **Turbines** installed between the two basins
 - **Sluice gates** for controlled water flow

3. Working Principle:



Step 1: Basin Charging (High Tide)

- During rising tide, seawater enters the upper basin.
- Water is stored until tide reaches maximum level.

Step 2: Power Generation (Controlled Discharge)

- Water is released from the upper basin to lower basin through turbines.
- The head difference between basins allows steady flow and turbine rotation, producing electricity.

Step 3: Low Tide Drainage

- As tide recedes, the lower basin discharges its water back to the sea.
 - The upper basin is ready to be refilled during the next high tide.
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7b. What are the advantages and limitations of tidal energy?

Advantages of Tidal Energy:

- 1. Renewable and Inexhaustible:** Tidal energy is derived from gravitational forces, making it naturally replenishable and endless.
- 2. Predictable Energy Source:** Tides follow a fixed and cyclic pattern, making tidal energy highly predictable and reliable.
- 3. Eco-Friendly:** No emission of greenhouse gases or pollutants, making it environmentally clean.
- 4. Long Lifespan:** Tidal power plants can function effectively for 50–100 years with minimal wear and tear.
- 5. Low Operating Costs:** After installation, maintenance and operational costs are low, ensuring cost-effectiveness in the long run.

Limitations of Tidal Energy:

- 1. High Initial Investment:** Construction of tidal barrages, turbines, and civil works involves very high capital costs.
 - 2. Limited Suitable Sites:** Effective tidal generation requires minimum tidal range of 5 meters, which is available only in a few coastal regions.
 - 3. Environmental and Ecological Impact:** Alteration in tidal flow may affect marine life, fish migration, and local ecosystems.
 - 4. Intermittent Generation:** Power generation happens only during tidal inflow and outflow periods, not continuously.
 - 5. Navigation and Sediment Issues:** Tidal structures may hinder boat movement and cause sediment accumulation.
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8a. With the neat sketch explain the working principle of OTEC power station. (Closed cycle) .

1. What is Closed Cycle OTEC?

- In a closed cycle Ocean Thermal Energy Conversion (OTEC) system, a low boiling point working fluid (like ammonia or propane) is used to drive the turbine.
- Ocean water is used only as a heat source and cooling medium, not as the working fluid.

2. Main Components:

- Warm surface seawater intake pipe
- Evaporator (heat exchanger)
- Turbine and generator
- Condenser (using cold deep seawater)
- Working fluid in a closed loop

3. Working Process (Step-by-Step):

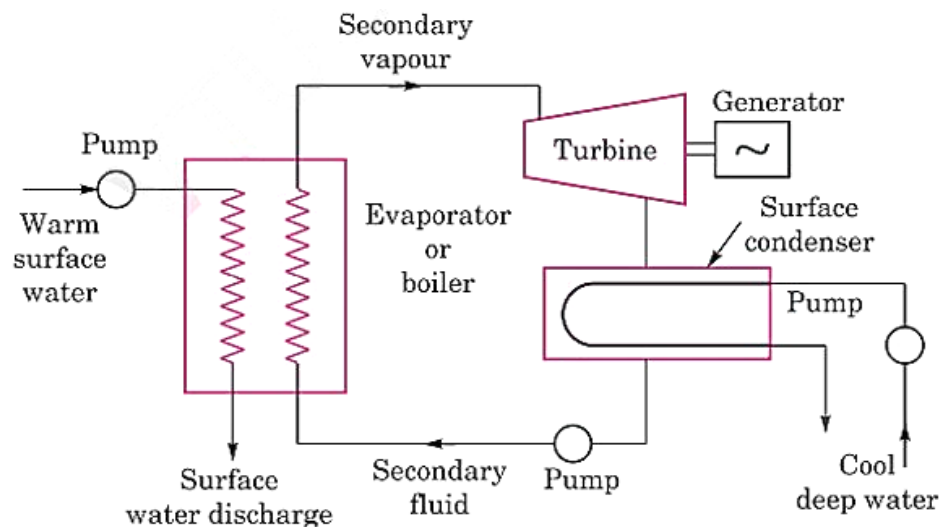


Fig 11: Schematic of Closed cycle OTEC system

1. Warm Water Heats Working Fluid

- Warm seawater (25–30°C) is pumped into a heat exchanger (evaporator).
- The heat is transferred to the working fluid (like ammonia), which boils due to its low boiling point.

2. Vapor Drives Turbine

- The vaporized working fluid expands and flows through a turbine, causing it to rotate.
- This turbine is connected to a generator, which produces electricity.

3. Condensation Using Cold Water

- After passing through the turbine, the vapor enters a condenser, where cold deep seawater (~5°C) is circulated.
- This cools and condenses the vapor back to liquid form.

4. Recirculation

- The liquid working fluid is pumped back to the evaporator, completing the closed loop cycle.

8b. What are the problems associated with OTEC.

1. High Initial Cost

- Construction of large underwater pipelines, platforms, and heat exchangers requires heavy investment.
- Not economically feasible for many countries without government support.

2. Low Thermal Efficiency

- Since the temperature difference between surface and deep water is small (about 20°C), the overall efficiency is very low ($\approx 2\text{--}3\%$).

3. Technological Challenges

- Handling corrosion, biofouling, and scaling in the marine environment is technically complex and requires advanced materials and coatings.

4. Environmental Concerns

- Discharge of cold deep water back into the ocean can disturb marine ecosystems, especially in sensitive regions.

5. Limited Suitable Locations

- OTEC plants can be set up only in tropical regions where surface water is warm and deep water is cold — so site selection is limited.

6. Transmission of Power

- If OTEC plant is offshore (floating), transmitting power to the land-based grid becomes difficult and expensive.

7. Large Size Requirements

- To make power generation meaningful, OTEC systems must be built on a large scale, requiring huge installations and sea space.
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MODULE 05

9a. Explain the biomass sources for biogas generation.

Biomass Sources for Biogas Generation

- Biogas is generated by anaerobic digestion of biomass materials.
- The following are the main sources used for biogas production:

1. Animal Waste: Animal dung such as cow dung, poultry litter, and piggery waste are the most commonly used biomass sources. These are easily available in rural areas and rich in organic matter.

2. Agricultural Residues: This includes crop waste like wheat straw, rice husk, corn stalks, and groundnut shells. Though dry, they can be mixed with dung or pre-treated to improve gas yield.

3. Kitchen and Food Waste: Vegetable peels, leftover food, and fruit waste from households, hotels, and restaurants are rich in moisture and nutrients, making them excellent for biogas generation.

4. Sewage and Sludge: Human waste from toilets and sewage treatment plants is already semi-liquid and easy to digest by bacteria in biogas systems.

5. Industrial Organic Waste: Waste from food processing, dairy industries, breweries, and sugar factories has a high gas production potential due to its organic content.

6. Aquatic Plants and Algae: Water hyacinth, duckweed, and algae grow rapidly and are suitable for use in regions where land-based biomass is limited.

7. Forestry Waste: Materials like sawdust, wood chips, bark, and dry leaves can be used after processing, though their digestion is slower.

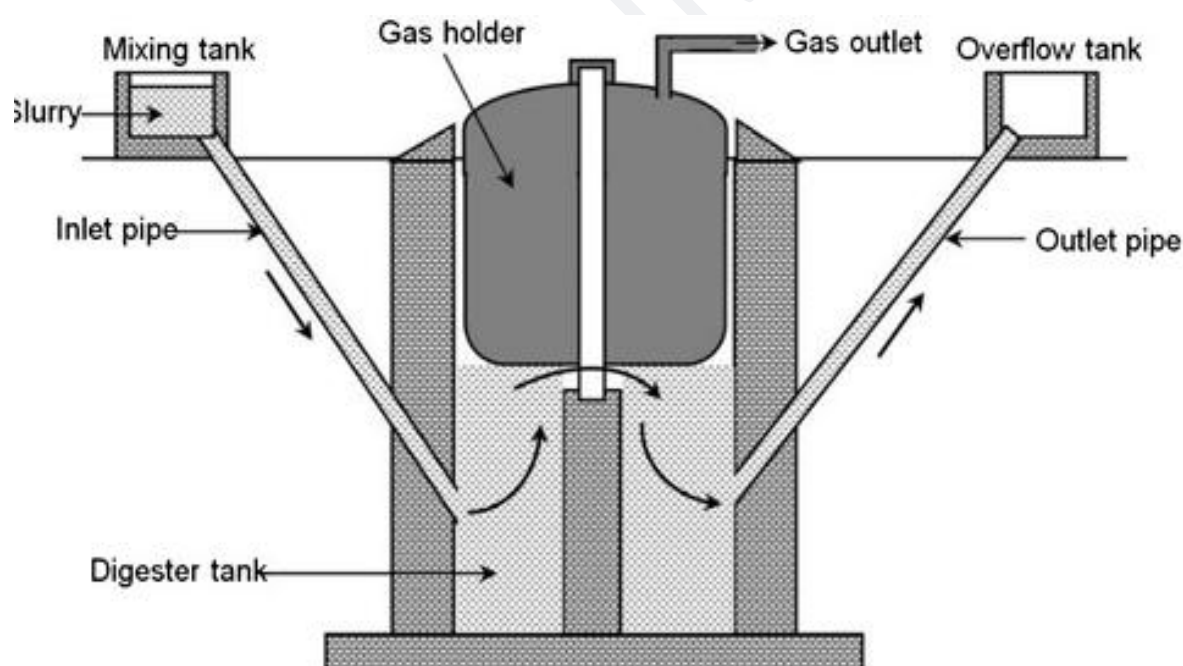
Conclusion: Biomass for biogas can come from various sources like animal dung, crop residues, kitchen waste, and industrial sludge. Utilizing these materials helps generate renewable energy while managing organic waste effectively.

9b. Sketch and explain the working of a floating dome type biogas plant used in India.

1. Introduction: The floating dome biogas plant, also called the KVIC model, is commonly used in India to convert organic waste like cow dung into useful biogas for cooking and lighting.

2. Main Components:

- **Digester Tank:** Underground sealed tank where anaerobic digestion of biomass occurs.
- **Inlet Chamber:** Used to feed the slurry (cow dung + water).
- **Outlet Chamber:** Removes digested slurry, which is used as organic fertilizer.
- **Floating Gas Holder (Drum):** Steel or plastic drum that floats on the slurry and stores biogas.
- **Gas Outlet Pipe:** Carries the gas to homes for use.



3. Working:

1. A mixture of cow dung and water (called slurry) is prepared and poured into the inlet chamber.
2. The slurry flows into the digester tank where it remains for several days.

3. Anaerobic bacteria decompose the organic matter and generate biogas.
 4. The biogas gets collected in the floating drum, which rises as gas accumulates.
 5. The drum maintains a constant gas pressure due to its weight.
 6. The gas is delivered to homes through the gas outlet pipe for cooking or lighting.
 7. The spent slurry flows into the outlet chamber and is periodically removed for use as organic manure.
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10a. Describe the main sources from which hydrogen can be obtained for energy use.

1. **Electrolysis of Water:** Electricity is used to split water into hydrogen and oxygen. If renewable energy is used, it's a clean method.
 2. **Steam Methane Reforming (Natural Gas):** Hydrogen is extracted from methane using steam. It is widely used but emits CO₂.
 3. **Coal Gasification:** Coal is reacted with steam and oxygen to produce a gas mix from which hydrogen is separated.
 4. **Biomass Gasification:** Organic waste or crop residues are heated to produce hydrogen-rich gases. It is renewable and eco-friendly.
 5. **Photobiological Processes:** Algae or bacteria release hydrogen under sunlight through biological reactions. It is still experimental.
 6. **Photoelectrochemical Water Splitting:** Sunlight is directly used with special materials to split water. It is a developing green technology.
 7. **Thermochemical Water Splitting:** High-temperature heat from solar or nuclear sources splits water into hydrogen and oxygen.
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10b. Explain the basic principle behind biochemical production of hydrogen.

Biochemical Production of Hydrogen – Basic Principle

Hydrogen can be produced biologically using microorganisms under specific conditions. This is a clean, sustainable, and eco-friendly method.

Basic Principles:

- 1. Microorganism-based Process:** Hydrogen is produced using bacteria or algae, which break down organic matter.
 - 2. Enzymatic Reaction:** Enzymes like hydrogenase and nitrogenase present in microbes help in releasing hydrogen during their metabolism.
 - 3. Organic Substrates Used:** Feedstock like agricultural waste, sewage, or sugars are used as raw materials.
 - 4. Anaerobic Conditions:** The process usually occurs in oxygen-free (anaerobic) conditions which are ideal for hydrogen production.
 - 5. Types of Fermentation:**
Hydrogen is produced through:
 - **Photo fermentation** (in presence of sunlight)
 - **Dark fermentation** (without light)
 - 6. Low Energy Input:** The process operates at low temperature and pressure, making it energy-efficient.
 - 7. Eco-Friendly Method:** No greenhouse gases are emitted; it is considered environmentally safe and renewable.
-